CSMFAB000000000113-Vol-09

Smart Muscle Spear Design Across All Five Scales

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1. Design Philosophy: The Smart Muscle Spear

The "Smart Muscle" designation reflects three integrated active functions: 1. **Archimedean screw muscle:** Rotating blades generate thrust in water (the primary "muscle") 2. **Piezoelectric flutter damping:** KNbO3-BaTiO3 patches damp aerodynamic vibration actively 3. **Adaptive payload shroud:** Deployable nose section protects payload during water transit

All sizes maintain geometric similarity for predictable scaling behavior.

1.1 Geometric Scaling Laws

For a family of geometrically similar spears (constant fineness ratio L/D = 8):

Scale	Length L (m)	Diameter D (m)	Screw Radius R (m)
Scale Factor 1 (Nano, 50g)	0.16	0.020	0.010
Scale Factor 2 (Micro, 1 kg)	0.32	0.040	0.020
Scale Factor 3 (Small, 10 kg)	0.64	0.080	0.040
Scale Factor 4 (Medium, 100 kg)	1.28	0.160	0.080
Scale Factor 5 (Large, 1000 kg)	2.56	0.320	0.160

1.2 Mass Properties (ZrB2-SiC, hollow shell design)

Solid cylinder mass: m = rho * pi * (D/2)^2 * L For ZrB2-SiC (rho=6090 kg/m3), hollow shell (wall thickness t = 0.1*D):

```
V_shell = pi * D * L * t = pi * D * L * 0.1*D = 0.1 * pi * D^2 * L
m_spear = rho * 0.1 * pi * D^2 * L = 6090 * 0.1 * pi * D^2 * L
```

Scale	D (m)	L (m)	m_spear (kg)	I_axial (kg · m2)	L_gyro @ 3000RPM
Nano	0.020	0.16	0.061	1.5e-6	4.7e-4
Micro	0.040	0.32	0.49	2.4e-5	7.5e-3
Small	0.080	0.64	3.90	3.8e-4	0.120
Medium	0.160	1.28	31.2	6.1e-3	1.91
Large	0.320	2.56	249	9.8e-2	30.7

1.3 KNbO3-BaTiO3 Flutter Damper Placement

Four patches per spear, at 1/4-chord from base:

Scale	Patch size (mm)	d33=285 pC/N	Damping ratio added
Nano	5x5x1	285 pC/N	+0.05 (passive shunt)
Micro	10x10x2	285 pC/N	+0.06
Small	20x20x3	285 pC/N	+0.06
Medium	40x40x5	285 pC/N	+0.07
Large	80x80x10	285 pC/N	+0.08

All scales achieve damping ratio zeta > 0.08 (from 0.02 undamped) — flutter suppressed.

2. Screw Blade Design

2.1 Blade Geometry

Each scale has 3 helical blades (empirically optimal for thrust/drag ratio): - Helix angle: 2.4° (from CSMFAB000000000109 optimal, independent of scale) - Chord-to-pitch ratio: 0.85 (Glauert optimum) - Blade tip-to-body radius ratio: 2.0 (R_blade = 2 * R_body)

2.2 Blade Stress Analysis

Critical: centrifugal stress at blade root during 3000 RPM spin:

```
sigma_centrifugal = 0.5 * rho_blade * omega^2 * (R_blade^2 - R_root^2)
                  = 0.5 * 6090 * (314)^2 * (4*R_body^2 - R_body^2)
                  = 0.5 * 6090 * 98,596 * 3 * R_body^2
                  = 900,000 * R_body^2   [Pa, if R in meters]
```

Scale	R_body (m)	sigma_centrifugal (MPa)	ZrB2-SiC limit (MPa)	Safety Factor
Nano	0.010	0.09	850	9,444
Micro	0.020	0.36	850	2,361
Small	0.040	1.44	850	590
Medium	0.080	5.76	850	147
Large	0.160	23.0	850	37

All scales have enormous safety factors. ZrB2-SiC is thermally and mechanically ideal.

3. Spin Motor: KNbO3-BaTiO3 Ultrasonic for All Scales

From CSMFAB000000000020: KNbO3-BaTiO3 ultrasonic motor at 200mm diameter → 500 N · m, 240 rad/s.

Scaling (torque scales as R^3 for geometrically similar motors):

Scale	Motor Size	Torque (N · m)	Power (kW)	Spin-Up Time
Nano	10mm dia	0.0025	0.8 W	0.6 s
Micro	20mm dia	0.02	6.3 W	1.2 s
Small	40mm dia	0.16	50 W	2.4 s
Medium	80mm dia	1.28	400 W	4.8 s
Large	160mm dia	10.2	3.2 kW	9.6 s

These motors are integrated into the spear core, spinning up to 3000 RPM before deployment.

Citations (Vol 09)

- CSMFAB000000000001 V2.0, ZrB2-SiC material specs, CSM 2026
- CSMFAB000000000020 V2.0, KNbO3-BaTiO3 motor design, CSM 2026
- CSMFAB000000000109 V2.0, Spear geometry, CSM 2026
- Etkin, B., Reid, L.D., Dynamics of Flight: Stability and Control, 3rd Ed, Wiley 1996
- McCormick, B.W., Aerodynamics, Aeronautics, and Flight Mechanics, 2nd Ed, Wiley 1994
- von Karman, T., Aerodynamics, Dover 2004 (scaling laws reference)

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